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A wish-list of general characteristics of intelligence

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- **Planning**: formulation of plans – sequences or agenda of actions to accomplish externally or internally determined goals

What is AI?

Thinking Humanly

"The exciting new effort to make computers think … *machines with minds*, in the full and literal sense." (Haugeland, 1985)

"[The automation of] activities that we associate with human thinking, activities such as decision-making, problem solving, learning ...." (Bellman, 1978)

Thinking Rationally

"The study of mental faculties through the use of computational models." (Charniak and McDermott, 1985)

"The study of computations that make it possible to perceive, reason, and act." (Winston, 1992)

Acting Humanly

"The art of creating machines that perform functions that require intelligence when performed by people." (Kurzweil, 1990)

"The study of how to make computers do things at which, at the moment, people are better." (Rich and Knight, 1991)

Acting Rationally

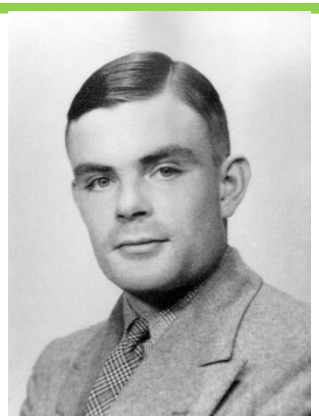
"Computational Intelligence is the study of the design of intelligent agents." (Poole *et al*, 1998)

"AI … is concerned with intelligent behavior in artifacts." (Nilsson, 1998)

Acting Humanly: Turing Test

Alan Turing (1950)

Operational definition of intelligence



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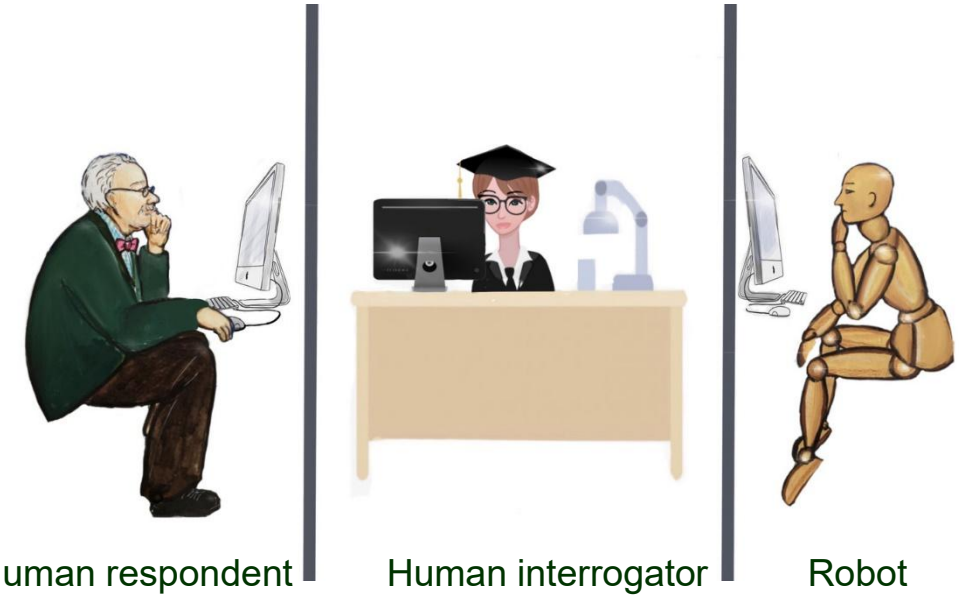

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- A human interrogator poses some written questions to another human and a computer (or “robot”).
- The computer passes the test if the interrogator cannot tell whether the written responses come from the human responder or the computer.

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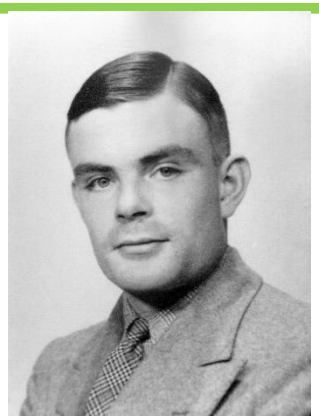

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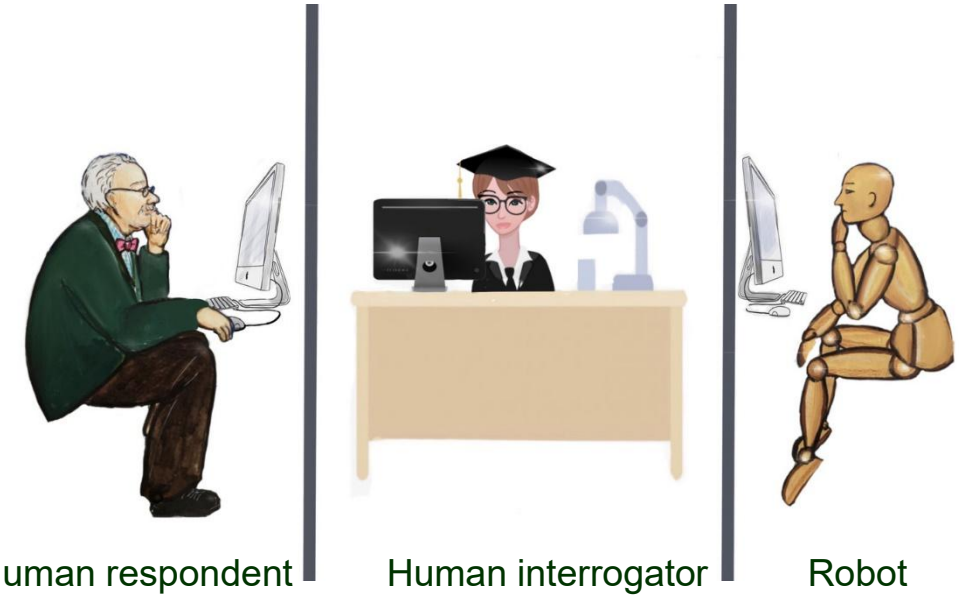

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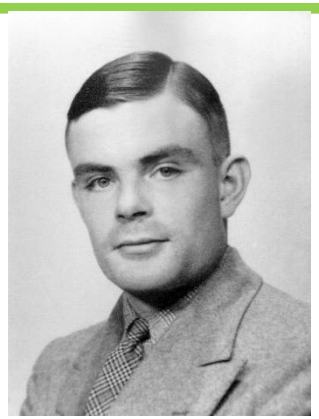

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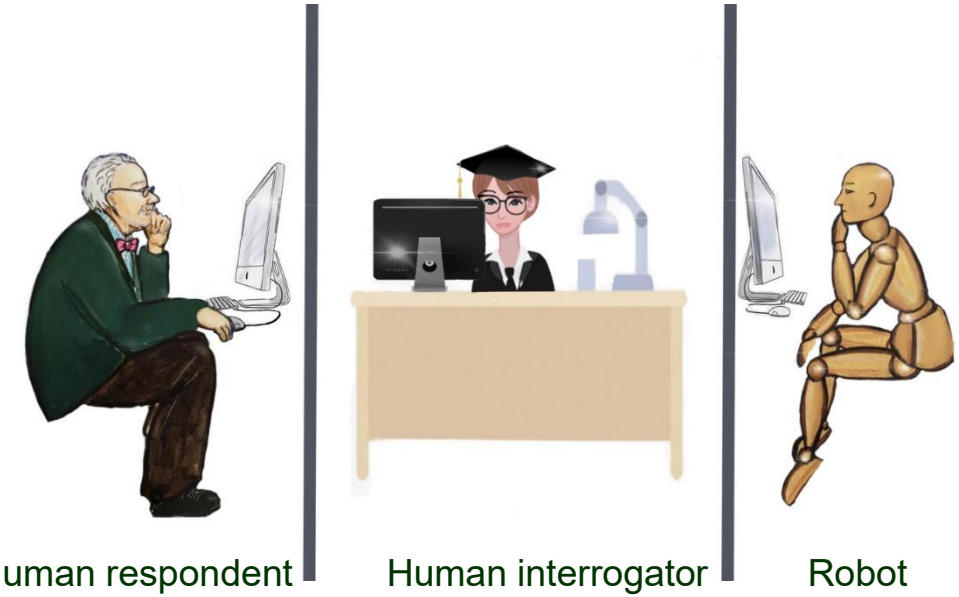

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ChatGPT 4.5 passed the Turing test? (judged to be the human 73% of the time)

Six Disciplines of AI

To pass the Turing test, the following capabilities are required:

- Natural language processing (NLP) to communicate in a human language;
- Knowledge representation to store what it knows;
- Automated reasoning to answer questions and draw new conclusions;
- Machine learning (ML) to adapt to new circumstances and to detect patterns.

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Most vibrant and in-demand areas in AI today: ML, CV, NLP, and Robotics.

Thinking Humanly: Cognitive Modeling

To say a program thinks like a human, we must know humans think first.

Learn about human thought in three ways:

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- psychological experiments – observing a person in action;
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Cognitive science constructs theories of the human mind by combining

- computer models from AI
- experimental techniques from psychology

Thinking Rationally: “Laws of Thought”

Represent problems using logic and build programs to create intelligent systems.

Circumscription by John McCarthy (1980)

Drawbacks:

- Not easy to state informal (and often uncertain) knowledge in the formal logical notation.
- Big difference between solving a problem “in principle” and solving it in practices.

Acting Rationally: Rational Agent

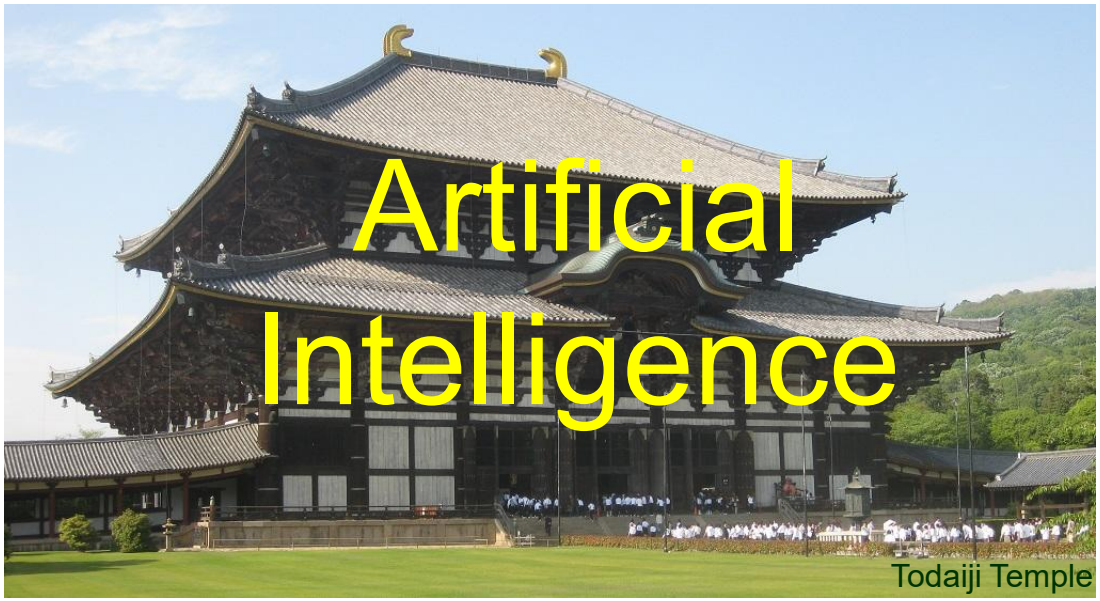
This course is about designing rational agents.

Rational behavior: doing the right thing

expected to maximize goal achievement,
given the available information

- An agent is an entity that perceives and acts.
- A rational agent is one that acts to achieve the best outcome.

Foundations of AI



Todaiji Temple

Philosophy

Mathematics

Economics

Neuralscience

Psychology

Computer Engineering

Control Theory

Linguistics